

# Balsa Glider Challenge

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Class: Period 7 Aerospace

1. The wing control panel is used to lay out the wing planform by adjusting the wing Span, Root Chord, Taper Ratio, and Leading Edge Sweep Angle. This panel is where you set the approximate velocity at which the glider will be launched from the catapult that will be used during the flight testing phase.
2. It is only showing the wings which control the direction of flight. This is okay because the wings are what ultimately make the glider fly regardless of the fuselage
3. You navigate using the multiple tabs on the top of the screen including file, main, wing, and preferences.
4. The position of the wings and the size and chord are the most important parts of the things that affect the performance of the glider, regardless of the fuselage size.

## Conclusion of Challenge 1

1. They were all easy to understand, however things like chord I did not understand. A chord is a principal member of a truss.
2. It would probably be an issue to get the exact values that I used
3. Trying to find out how to edit the thing was very difficult and required a lot of trial and error.

## Conclusion of Challenge 2

1. I actually liked this one a lot more because it was a cooler design in my opinion. I kept trying to move the components but had trouble doing it.
2. The stabilizer and the wing were literally flipped, so it presented a whole other type of problem.

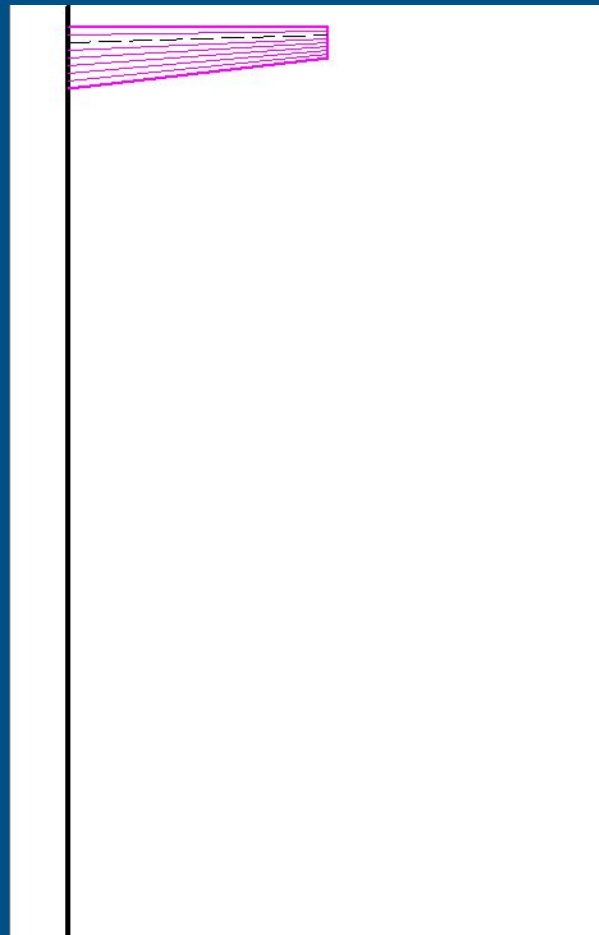
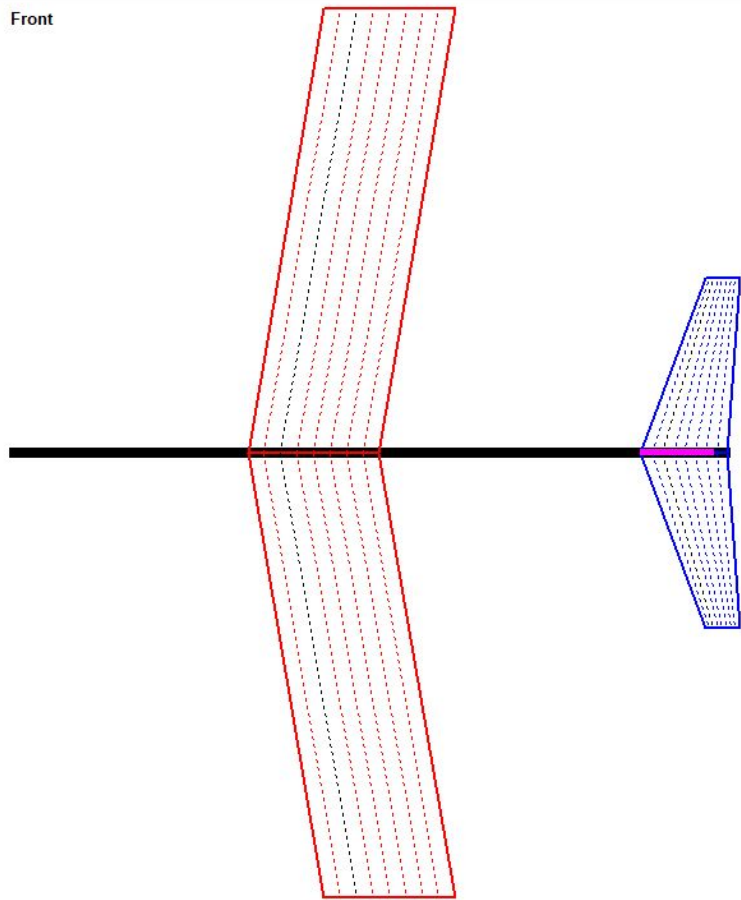
| Main        | Wing  | Stabilizer | Vertical Tail | Information                 |                             |
|-------------|-------|------------|---------------|-----------------------------|-----------------------------|
| <div></div> | 41.51 |            |               | Fuselage Length (cm)        | Glider CG 0018.99 cm        |
| <div></div> | 13.69 |            |               | Wing Location (cm)          | Glider Mass 0038.93 g       |
| <div></div> | 36.45 |            |               | Stabilizer Location (cm)    | Wing Loading 0000.10 g/cm^2 |
| <div></div> | 36.53 |            |               | Vertical Tail Location (cm) |                             |
| <div></div> | 7.41  |            |               | Nose Mass (g)               |                             |

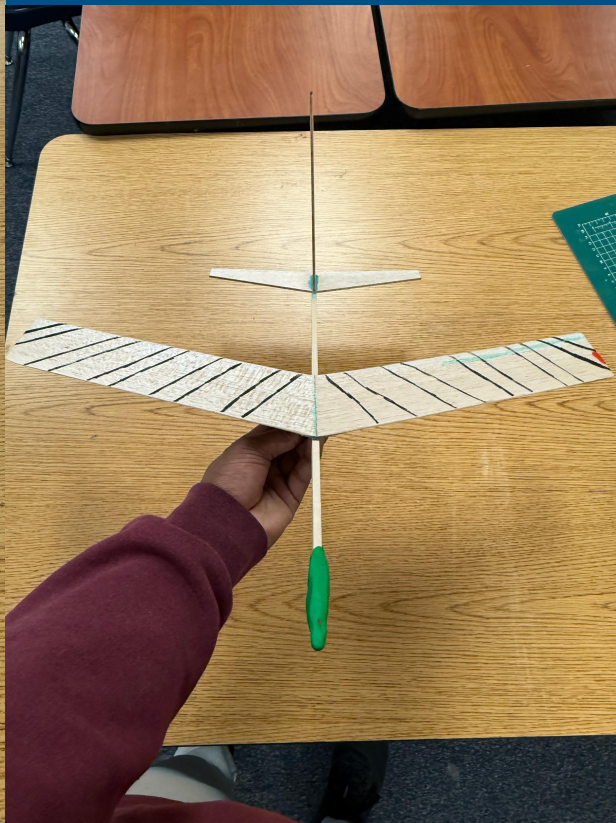
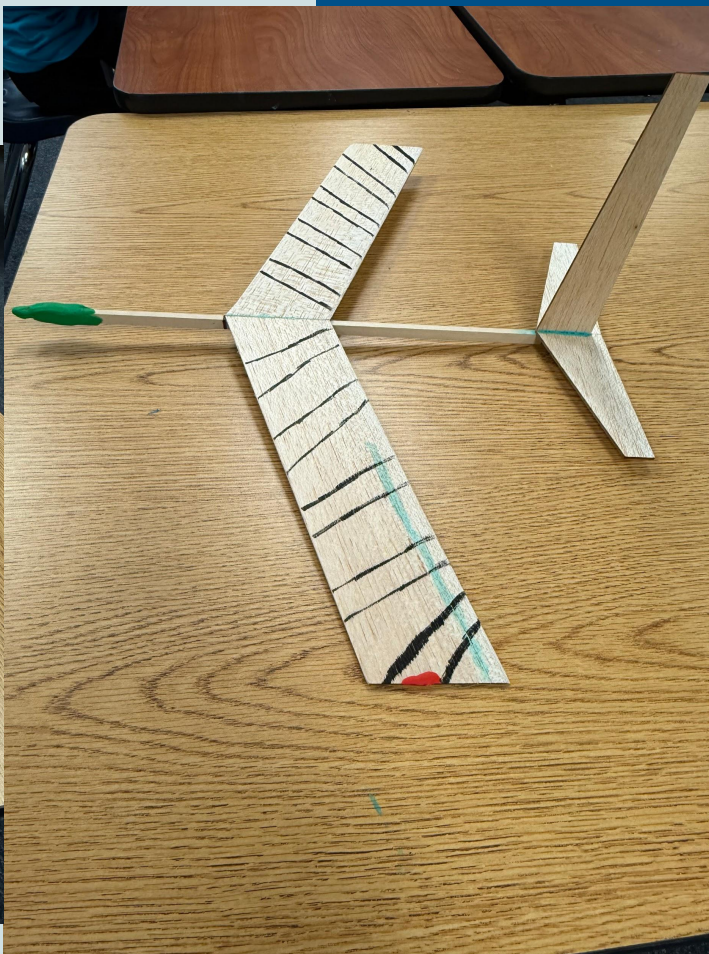
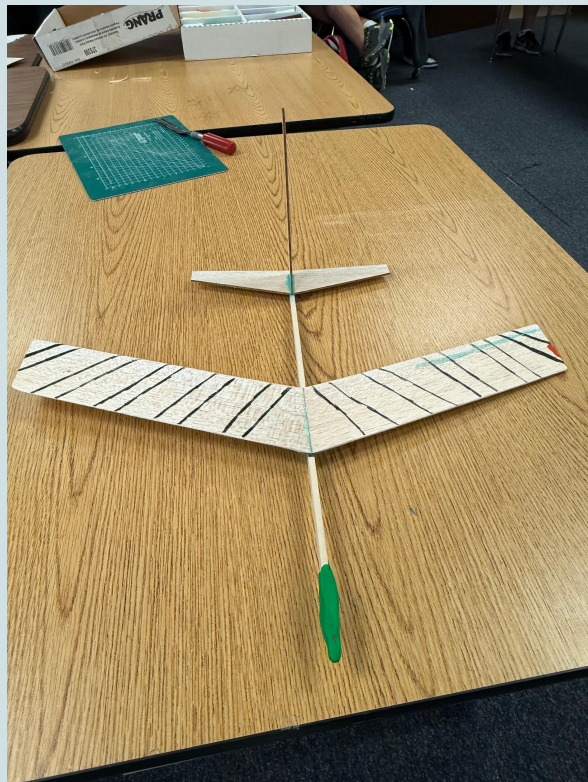
| Main        | Wing  | Stabilizer | Vertical Tail | Information              |                            |
|-------------|-------|------------|---------------|--------------------------|----------------------------|
| <div></div> | 51.47 |            |               | Span (cm)                | Planform Area 0391.17 cm^2 |
| <div></div> | 7.60  |            |               | Root Chord (cm)          | Aspect Ratio 0006.77       |
| <div></div> | 1.00  |            |               | Taper Ratio              | Tip Chord 0007.60 cm       |
| <div></div> | 9.76  |            |               | Leading Edge Sweep Angle | Tip Sweep 0004.43 cm       |
| <div></div> | 24.01 |            |               | Velocity (km/hr)         | Arbitrary Sweep Angle      |

| Main        | Wing  | Stabilizer | Vertical Tail | Information              |                            |
|-------------|-------|------------|---------------|--------------------------|----------------------------|
| <div></div> | 20.24 |            |               | Span (cm)                | Planform Area 0071.12 cm^2 |
| <div></div> | 5.02  |            |               | Root Chord (cm)          | Aspect Ratio 0005.76       |
| <div></div> | 0.40  |            |               | Taper Ratio              | Tip Chord 0002.01 cm       |
| <div></div> | 20.12 |            |               | Leading Edge Sweep Angle | Tip Sweep 0003.71 cm       |
| <div></div> |       |            |               |                          | Arbitrary Sweep Angle      |

| Main        | Wing  | Stabilizer | Vertical Tail | Information              |                            |
|-------------|-------|------------|---------------|--------------------------|----------------------------|
| <div></div> | 17.76 |            |               | Height (cm)              | Planform Area 0054.78 cm^2 |
| <div></div> | 4.14  |            |               | Root Chord (cm)          | Aspect Ratio 0005.76       |
| <div></div> | 0.49  |            |               | Taper Ratio              | Tip Chord 0002.03 cm       |
| <div></div> | 0.00  |            |               | Leading Edge Sweep Angle | Tip Sweep 0000.00 cm       |
| <div></div> |       |            |               |                          | Arbitrary Sweep Angle      |

Front







## Flight Data Recording & Design Evaluation

Glider Type  
(Typical or Canard)

Typical

Designer  
Name:

Rhithvik Selvakumar

Evaluated  
by

Reed Moore

### Summary Data: Glider Design and Construction

#### Aerodynamic Characteristics

Main Wing and Stabilizer (shape, symmetry, alignment)

Leading and Trailing Edges (rounding, taper, straight)

Angles of Incidence (match to plans)

2/3 are 4's (Bonus of 1)

2/3 are 5's (Bonus of 2)

#### Appearance and Durability

Construction (neatness, accuracy, gaps)

Decorations (neatness, creativity, appropriateness)

Durability (due to construction quality)

2/3 are 4's (Bonus of 1)

2/3 are 5's (Bonus of 2)

**Sub-Total:** (34 possible)

Designer  
*Enter a rubric score*

Evaluator  
*Enter a rubric score*

|   |   |
|---|---|
| 5 | 5 |
| 4 | 4 |
| 5 | 5 |
|   |   |
| 2 | 2 |

*Enter a rubric score*

*Enter a rubric score*

|    |    |
|----|----|
| 5  | 5  |
| 3  | 2  |
| 5  | 5  |
|    |    |
| 2  | 2  |
|    |    |
| 31 | 30 |

### Individual Event Data: Flight Performance (Launch Velocity $L = 24\text{km/hr}$ to match AERY design)

| Flight | Launch Angle (deg) | Launch Velocity (A-Q) | Time Aloft (s) | Distance (feet) | Velocity (ft/s) | Pitch Stability | Directional Stability | Changes Made                                |
|--------|--------------------|-----------------------|----------------|-----------------|-----------------|-----------------|-----------------------|---------------------------------------------|
| 1      | 10                 | M                     | 1.87           | 37              | 19.79           | NA              | Right                 | change shape of nose                        |
| 2      | 10                 | L                     | .88            | 34              | 38.64           | na              | na                    | change angle and increase start <u>velo</u> |
| 3      | 15                 | O                     | 1.34           | 36              | 26.87           | na              | na                    | increase start <u>velo</u>                  |
| 4      | 15                 | N                     | 1.93           | 40              | 20.73           | na              | na                    | na                                          |
| 5      | 20                 | N                     | 2.1            | 45              | 21.43           | na              | na                    | up angle                                    |
| 6      | 25                 | L                     | 2.2            | 35              | 15.91           | no              | na                    | more velo                                   |
| 7      | 25                 | M                     | 1.5            | 16              | 10.67           | went up         | na                    | lower angle                                 |
| 8      | 10                 | R                     | 2.4            | 42              | 21.67           | went up         | na                    | lower angle                                 |
| 9      | 0                  | R                     | 3.1            | 53              | 17.10           | na              | na                    | na                                          |
| 10     | 0                  | R                     | 2.7            | 48              | 17.78           | na              | na                    | na                                          |

## Competition Data: Flight Performance

| Flight Number | Distance (feet)                                       | Pitch Stability | Directional Stability |
|---------------|-------------------------------------------------------|-----------------|-----------------------|
| 1             | 38                                                    | 5               | 5                     |
| 2             | 58                                                    | 5               | 5                     |
| 3             | 69(max distance available hit door at end of hallway) | 5               | 5                     |

| Ave Distance | Ave Pitch | Ave Directional |
|--------------|-----------|-----------------|
| 55           | 15        | 15              |

### Flight Characteristics

Distance (Straight line, <24', 24-36', >36')

Pitch Stability (Uniform descent, gentle pitching, drastic)

Directional Stability (Straight, gentle turn, sharp curve)

2/3 are 4's (Bonus of 1)

2/3 are 5's (Bonus of 2)

**Total Score:** (17 Possible)

|                      |                                  |
|----------------------|----------------------------------|
| <b>DESIGNER</b>      |                                  |
| Enter a rubric score |                                  |
| 5                    | <input type="button" value="v"/> |
| 5                    |                                  |
| 5                    |                                  |
|                      |                                  |
| 2                    |                                  |
| 17                   |                                  |



1. Was the glider as stable as you expected? Why or why not might this be so?

After construction my glider was tipping to the right which I believe was the result of uneven sanding. To counter the tilt I added a small amount of putty to balance my plane on the launcher handle. This change allowed my glider to be far more stable during flight than what I originally expected.

2. What techniques did you use to "trim" the glider for straight and gently descending flight?

Other than correcting my tilt, start velocity, and launch angle there was no other trim.

3. How many test flights were required to get the glider trimmed for long distance, straight-line flight? Was this expected? Why or why not?

It took about 9 flight trials to get the glider trimmed. This was expected as this was the amount of trials to get the correct amount of putty on the underweight wing, find the optimal launch angle, and find the optimal launch velocity.

4. Do you think the success or failure of your flight was due to the design, the construction or some other factor? Why?

I believe that the success of my flight was due to the construction as the shaping of my wing camber during the sanding process went really well.

5. What do you consider as successful with your glider

My glider is considered successful because it had a stable flight and maxed out the distance it could go in the hallway during testing and competition.

6. What would you do differently if you could design a 2nd glider?

Nothing other than making sure I sand evenly